Coherent-Offset Aliasing in UAV-to-Satellite Geo-Localization: Taxonomy, Measurement, and Why Consistency-Based Rejection Fails

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***Abstract*—UAV geo-localization against satellite reference imagery fails on repetitive terrain in ways standard robustness tools cannot see. We characterize the failure on UAV-VisLoc and separate it into two classes: *whole-tile* aliases, where the matcher locks onto a look-alike tile hundreds of metres away, and *sub-tile* aliases, where the match lands on the correct tile but displaced by one period of a repeating structure invisible at satellite resolution. Three measurements make the case. First, signed per-frame offsets ($n = 610$ contiguous frames) show that magnitude histograms—the standard reporting convention—hide the alias class's bimodality and its hole at zero. Second, a benchmark of four published rejection designs over one accepted-fix stream shows that sequential consistency and ORB-SLAM's three-keyframe rule discriminate *backwards* on the sub-tile class (retaining wrong fixes at a higher rate than correct ones), while pairwise consistency and robust frame alignment discriminate forward only by keeping 7–40% of correct fixes; no consistency-based method meets a pre-registered adoption bar, whereas a prior-ratio gate separates the whole-tile class completely (0 of 7 fatal fixes kept per drift at 150 m and 300 m). Third, the alias offset is temporally coherent (2.55$\times$ below a shuffled null at lag 1)—which is exactly why consistency fails—and the mechanism-motivated countermeasure, a multi-hypothesis mixture filter, recovers none of the oracle gap because the alias lattice is per-field and unobservable in the satellite texture. The signature and rate structure reproduce under three further matchers (ORB, SIFT, SuperPoint+LightGlue); a dense matcher (RoMa) adds a third failure mode—a globally smooth, confident warp displaced 55 m–11 km—while the same family reports 15 m at 95% on contemporary matched-modal imagery, showing published dense-matcher accuracies presume modal proximity. Ten-match averaging cuts the pooled median 21.8 to 8.0 m—except on whole-tile terrain, where short averages first do worse (Finding 12). A spacing-isolation experiment on 429 video-rate frames identifies temporal spacing as the governing variable: geo-registered fix propagation converts 74% per-frame solvability into 97% frame coverage at zero accuracy cost, while the identical chain at 250 m stride collapses to 25%. A boundary map over all eleven regions, an executed replication on an independent urban dataset (which also exposes a 16.5 m georeferencing bias), and a pre-registered replication protocol complete the contribution.**

***Index Terms*—Geo-localization, perceptual aliasing, robust estimation, UAV, satellite imagery, failure analysis.**

I. Introduction

GPS-denied UAV navigation increasingly relies on matching nadir drone imagery against georeferenced satellite tiles to anchor drift-prone visual-inertial odometry. Published systems report median errors of metres on favourable terrain [1]–[3]. Failure analysis, however, is thin: benchmarks report accuracy at thresholds (A@X m) over pooled terrain, which is blind to how, where, and why the tail fails. Recent mitigation systems acknowledge aliasing explicitly—trajectory-level VPR fusion [4], particle filtering with learned aliasing gates [5], correspondence propagation [2]—but they assume the temporal-consistency and trusted-odometry regime they need; the behaviour of that regime's rejection tools on the alias classes themselves has not been measured.

We document a failure class on repetitive terrain—farmland with parallel furrows, forest canopy—where accepted matches are wrong by tens to hundreds of metres *with full geometric and appearance support*. We show this class defeats the standard robustness toolbox for a measurable, mechanical reason, we show the failure is not an artifact of any one matcher—sparse or dense—and we quantify what would be required to fix it, including the one regime change (video-rate capture with fix propagation) that measurably contains it.

Coherent (mutually consistent) outliers in SLAM are studied theoretically for indoor scenes [6]; row-level aliasing on vineyard rows is reported from ground robots with LiDAR [7]–[9]. Neither provides cross-view (nadir UAV $\leftrightarrow$ satellite) measurements, per-class rejection behaviour, nor the temporal-coherence and appearance-optimality measurements we report. STHN notes self-similar patterns as a confound in UAV thermal$\leftrightarrow$satellite homography [10] and UASTHN detects such failures via uncertainty estimation [11], but neither analyzes the alias structure. Robust pose-graph back-ends (PCM, GNC, TACO) are designed against exactly this outlier shape [12]–[15]; we measure how the standard representatives of this family behave on real cross-view data. Learned dense matchers (RoMa, LoFTR, RoMav2) are benchmarked on matched-modal contemporary imagery [16]; we measure the same family on vintage-mismatched pairs and show the benchmark numbers do not transfer—a finding consistent with concurrent work that documents dense-matching unreliability under temporal gaps at remote-sensing scale [17] and adapts RoMa specifically to survive multi-temporal UAV-to-orthophoto pairs [18].

Contributions.

1) **A taxonomy** separating whole-tile from sub-tile aliases, each with a distinct geometry, temporal signature, and

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rejection behaviour (Sec. IV-A, IV-C).

2) **Backwards-rate tables** for four published rejection methods over a shared fix stream: sequential consistency and the three-keyframe rule keep wrong fixes at a *higher* rate than correct ones; PCM and robust frame alignment discriminate forward only by destroying 60–93% of correct fixes; only a prior-normalized distance gate works, and only on the whole-tile class (Sec. IV-B).
3) **Matcher independence**: the alias signature and the rate structure reproduce under ORB-only, SIFT-only, and SuperPoint+LightGlue—three correspondence sources spanning binary corners, gradient blobs, and a learned detector–matcher—and dense matchers add a third failure mode with its own mechanism (Sec. IV-G).
4) **The sign-folding diagnostic**: absolute-value error histograms fold symmetric alias structure into an apparently continuous distribution; signed offsets recover the axis and the hole at zero (Sec. IV-A).
5) **A falsified countermeasure**: the mixture hedge filter that follows from the mechanism recovers none of the oracle gap, because the alias lattice is per-field and unobservable—with a reproducible oracle bound of 14.0 m median quantifying the residual (Sec. IV-D).
6) **An exhaustive terrain boundary map** over all eleven source-dataset regions plus held-out and cross-dataset controls confirming the class is terrain-specific, and a pre-registered replication protocol (Sec. IV-E, IV-F).
7) **A spacing-isolation measurement** (Sec. IV-I): at metre-scale frame spacing, geo-registered fix propagation—the mechanism underlying [2], [4]—carries one anchored fix for an entire flight at baseline accuracy (74%→97% frame coverage, zero accuracy cost); at 250 m stride the identical chain collapses. Temporal spacing—not the matcher, not the geometry—is the governing variable for fix propagation, with matched-modal controls showing terrain class operates independently of imagery vintage.

II. Related Work

Retrieval-based systems [1], [19], [20] and geometric pipelines (homography, as in the production matcher studied here; PnP against DSM [2], [3]) define **UAV–satellite geolocalization**, benchmarked by pooled accuracy-at-threshold. AnyVisLoc's Table 6 is representative: its baseline scores 74.1% A@5 m against an aerial photogrammetry reference but 18.5% A@5 m against a satellite reference [1]—the satellite-reference tail exists, is large, and is not analyzed per-region. Our work is complementary: we analyze *why* that tail exists rather than improving its head.

Perceptual aliasing and robust estimation. Aliasing in place recognition is a classic concern [21]; sequence-based VPR exploits temporal consistency to combat it [22]. Modern robust back-ends address mutually consistent outliers with pairwise consistency (PCM [12], Kimera-RPGO [13]), graduated non-convexity (GNC [14]), or robust kernels on pose-graph factors (TACO [15]). Lajoie et al. analyze coherent outliers in indoor SLAM theoretically and note that mutually consistent outliers defeat initial-guess-dependent rejection [6]. Vineyard robotics reports row-level aliasing from ground LiDAR qualitatively [7], [8], [23]. Multi-hypothesis handling of aliasing dates to hypothesize-and-verify loop closure [24]. None of these measures the rejection *rate* of published methods on a shared cross-view fix stream, nor the temporal coherence of the alias offset itself—the measurements we contribute.

Systems that assume the answer. A 2026 generation of deployed-system work builds the mitigation layer whose assumptions we test. NaviLoc [4] fuses per-frame VPR with VIO through a three-stage trajectory estimator (global SE(2) alignment, sliding-window Procrustes refinement, convex MAP smoothing with outlier clamping), reporting 19.5 m mean error on a rural benchmark at 9 FPS on an embedded board—its design presumes video-rate trusted relative motion and treatable VPR outliers. CPFL [5] runs a vision-only particle filter whose dual-granularity gating is explicitly designed "to mitigate perceptual aliasing," and contributes a Trajectory Continuity Index for continuous localization—an independent sign that the field's pooled per-frame metrics are felt to be insufficient (our signed-offset diagnostic and truth-correlation guard address the same reporting gap at the measurement level). OrthoTrack [2] propagates map-anchored correspondences to intermediate frames with optical flow—the chain architecture our Sec. III-E instantiates—and SeqLoc [25] aggregates sequential belief volumes for feature-sparse cross-view localization. All four operate in, or implicitly assume, the video-rate regime; none isolates frame spacing as a variable, and none measures rejection behaviour on coherent aliases. Our spacing-isolation pair (Sec. IV-I) and backwards-rate tables (Sec. IV-B) are the controlled measurements underneath this system family: they say when the assumed regime is necessary (250 m stride collapses the identical chain) and which of its rejection tools carry aliasing risk (consistency filters fail backwards; prior-ratio gates the separable class).

Error distribution analysis in VPR. Reporting conventions use magnitude histograms and CDFs; CPFL's continuity index is a trajectory-level addition. We show a class of symmetric bimodal error—the signature of period-quantized aliasing—is invisible in magnitude form, and propose signed-offset reporting as a diagnostic (Sec. IV-A).

Matcher dependence of failure rates. Failure analyses of matching pipelines are usually reported for the authors' own matcher; whether the reported tail is a property of the terrain or of the descriptor pipeline is rarely tested. We test it directly (Sec. IV-G): the alias signature is reproduced by every matcher family we can run, and the rejection-rate structure is stable in sign and magnitude across the classical matchers, with the learned matcher underpowered for rates but consistent at the signature level. Dense matchers (RoMa [26], RoMav2 [16]) report the strongest fine-localization numbers in the field; we show those numbers presume contemporary matched-modal imagery and that the family adds a distinct failure mode under vintage-mismatch. Concurrent work corroborates the regime dependence from the construction side: LoRetta [17] reformulates dense matching as localization-and-registration precisely because "intrinsically unmatchable regions make direct dense correspondence prediction unreliable" across acquisition-time

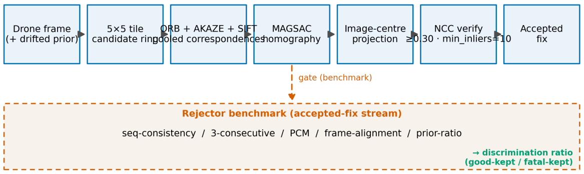

Fig. 1. The production map matcher under study and the measurement harness around it. Pooled ORB+AKAZE+SIFT correspondences feed a MAGSAC homography against a prior-centred satellite tile ring; accepted fixes carry the estimate, the prior, and the filter-reported uncertainty that the rejection benchmark consumes.

and season gaps, and Sat-RoMa [18] re-engineers RoMa with a satellite-tuned DINOv3 encoder to survive multi-temporal UAV-to-orthophoto pairs. Both build around the failure our Sec. IV-G isolates mechanistically in the stock models; neither characterizes the failure mode itself.

Fix propagation and odometry regimes. Tracking geo-registered control points between matched frames so that absolute matching becomes a rare re-anchor rather than a per-frame duty is standard in fielded systems [2], [27]. Whether the technique survives coarse frame spacing has not been isolated as a variable; the spacing-isolation experiment in Sec. IV-I is the first controlled measurement of the regime boundary for geometric fix propagation (SeqLoc's sequence aggregation is the retrieval-side analogue and reports the same single-frame collapse qualitatively).

III. Method

A. System under study

The pipeline is a production map matcher combining pooled correspondences from ORB, AKAZE and SIFT, MAGSAC homography [28], image-centre projection through the inverse homography, patch-wide NCC verification at 0.30 with the inlier floor at 10, DEM-corrected AGL scale. Candidate generation is prior-centred (5×5 tile ring). Frames carry a drifted prior injected as a random walk at 150/300/600 m. Dataset: UAV-VisLoc [19], eleven Chinese sites; all measurements use the dataset's drone GPS as truth.

B. Measurement protocol and definitions

Error classes. A fix is *good* if its error against the dataset GPS is $<$20 m, *mid* if 20–50 m, and *fatal* if $\geq$50 m. These boundaries are fixed a priori and used across all experiments.

Signed offsets. For each frame, we match against the ground-truth tile and record the signed (north, east) offset between the recovered position and the dataset's GPS record. Frames are *contiguous* in flight order ($n = 610$ solved on R04 out of 738 attempted), because step-sampling destroys the temporal structure the coherence question needs.

Fix stream. For rejection benchmarks, we run the production matcher under injected drift and keep every accepted fix with its estimate, prior, and prior uncertainty ($n = 40$ per region per drift, seed fixed).

Metrics. Good-kept% / fatal-kept% with denominators inline; discrimination ratio = good-kept/fatal-kept (>1 = forward discrimination). Adoption bar for a rejector: fatal cut $\geq$25% while keeping $\geq$80% of good fixes (pre-registered before any benchmark ran).

C. The rejector benchmark

Four published designs are implemented over the shared fix stream, using only deployed signals (priors, estimates, filter-reported uncertainty): sequential consistency (a fix survives if a neighbouring fix agrees within a motion-compensated tolerance; the ORB-SLAM3 covisibility principle [29]), ORB-SLAM's three-consecutive-keyframe rule [30], PCM-style greedy maximum clique [12], and VINS-Mono-style robust frame alignment (median prior$\leftrightarrow$estimate offset, reject residual $> \tau$) [31]. The prior-ratio gate (fix kept iff distance to prior $\leq 1.5\times$ the prior's uncertainty) is evaluated in deployed (filter RMS) and oracle (true prior error) forms.

D. Matcher variants

Every pooled-pipeline measurement has a variant protocol in which only the correspondence source changes, everything else (CLAHE preprocessing, DEM/AGL-corrected GSD rescale, GT tile patch, MAGSAC, thresholds) held identical: *ORB-only* (binary corner descriptors [32]), *SIFT-only* (gradient-histogram descriptors [33]), and *SuperPoint+LightGlue* (learned detector and matcher [34], [35]). Variant streams are analysed separately and never pooled with the production stream (cross-quote rule); the production matcher is the ORB+AKAZE+SIFT pool unless a table states otherwise. The dense variant (Sec. IV-G) is RoMa `roma_outdoor` [26] run on the same GT-tile protocol.

E. The propagation chain

A geo-registered fix-propagation chain in the family of OrthoTrack's map-anchored correspondence propagation [2], requiring no camera intrinsics: (i) *seed*: production pool match on a satellite crop centred at the known position (the flight-start GPS assumption); the seed homography maps drone pixels to map pixels. (ii) *chain*: KLT optical flow tracks the seed keypoints frame-to-frame; a MAGSAC relative homography is fit on the tracked pairs and composed with the accumulated absolute homography; position = frame centre projected through the composed map. (iii) *re-anchor*: if tracked points fall below 20, relative inliers below 10, or the propagated position exits a 90 m safe zone around the reference crop, the chain re-matches on a fresh crop centred at the last fix; on failure the frame coasts (no fix). The chain is evaluated on a video-rate dataset (XIAN-VisLoc [16], trajectory 16: 429 frames, 490×490, 3.8 m median frame step, Level-19 Google reference at 0.24 m/px, per-frame GPS) in two arms: dense (stride 1) and coarse (stride 65 $\approx$250 m—the source dataset's regime). The chain is deliberately minimal—no intrinsics, no learned components—because the experiment's object is the spacing variable, not the estimator; the contribution over [2] is the controlled stride pair, not the architecture.

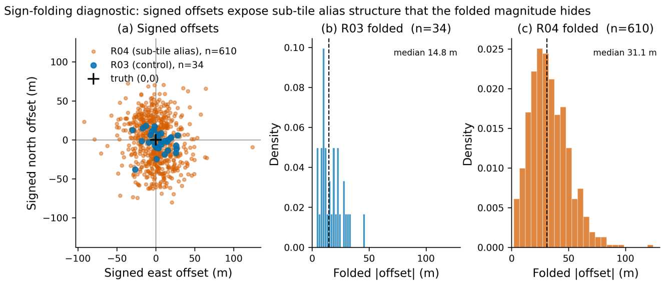

Fig. 2. The sign-folding diagnostic. Left: signed (north, east) offsets on R04 ($n = 610$ contiguous) and R03 (step-sampled). The R04 alias class is axis-aligned with a hole at zero; R03 is unstructured. Right: the same R04 data folded to magnitudes—the standard reporting convention—appears continuous and hides the alias class.

IV. Experiments

A. Error structure: the taxonomy and the sign-folding diagnostic

Setup. Signed offsets on GT tiles, R04 ($n = 32$ step-sampled for regional statistics; $n = 610$ contiguous for the full distribution) and R03 ($n = 30/557$) as control. The $n = 32$ sample and the $n = 610$ stream are *different* samples of the same region; baseline medians therefore differ (32.7 m on $n = 32$, 31.1 m on $n = 610$) and are always quoted with their sample size.

Result 1—magnitude histograms hide the alias class. The signed R04 offsets are axis-aligned and bimodal: axial resultant $R = 0.50$ (vs. 0.19 control) with the *hole at zero* (6/32 frames within ± 10 m along the dominant axis, vs. 14/30 on the control). Folded to magnitudes, this is exactly the "continuous, no-gap" distribution that earlier analysis misread as imprecision (Fig. 2).

Result 2—the axis is per-field, not per-region. At $n = 32$ the offsets split into two orientation groups (bearings 145–193° and 279–357°): the flight crosses fields whose furrow orientation changes. A single-region-axis oracle degrades accordingly: the sub-tile snap oracle (best integer period removed along one axis) falls from $2.39\times$ over a random-axis null at $n = 16$ to $1.72\times$ at $n = 32$, while its absolute ceiling stays stable (12.0–12.7 m median against a 32.7 m baseline). The alias is a lattice structure *per field*.

Result 3—the wrong lock is the appearance optimum, unanimously. At $n = 32$, masked patch NCC over displaced homographies selects displacement $k = 0$ on 32/32 frames; the oracle displacement recovers 20.4 m of median error (32.7→12.3 m). Appearance is not merely uninformative on this class; it is monotone *against* the truth.

Result 4—whole-tile aliases are a different class. On forest (R06), the error distribution is bimodal with a clean 40–150 m gap: discrete jumps to look-alike tiles 1–2 tile widths along track. Their offsets are coherent across the flight turn, following the aircraft.

Result 5—the signature replicates within the flight (split-half). The contiguous $n = 610$ stream split at its midpoint shows the signature independently in both halves: axis 168.1° vs. 166.6°, hole-at-zero 0.22 vs. 0.14, anisotropy 1.69 vs. 2.02, alias-group lag-1 coherence $1.16\times$ ($n = 13$) and $3.16\times$ ($n = 12$) below null, good groups at null. The weaker first-half coherence is consistent with Result 2—a half-flight still spans multiple fields with different axes.

TABLE I
Rejection rates over the shared accepted-fix stream (good-kept%/fatal-kept%; discrimination ratio >1 is forward). tol=τ=100 m throughout; prior-ratio threshold r=1.5; bold cells pass the pre-registered adoption bar (fatal cut $\geq$25%, good kept $\geq$80%).

Rejector	d150	d300	d600
Seq. consistency	0.96 (96/100)	0.84 (84/100)	0.81 (59/73)
3-consecutive	0.95	0.97	0.45 (n=3)
PCM max-clique	3.02 (40/13)	3.17 (23/7)	0.79
Frame alignment	2.09 (56/27)	2.80 (20/7)	1.12
Prior-ratio (RMS)	**2.12** (99/47)	1.08	1.00
Prior-ratio (oracle)	**2.31** (92/40)	**1.95** (97/50)	**1.57** (100/64)

Result 6—attitude and imagery vintage do not explain the class. Two reviewer-grade confounds are ruled out with the dataset's own metadata. (a) *Camera attitude.* The stream is effectively nadir: 608/610 frames have $|\text{tilt}| \leq 10°$ (median $|\text{pitch}|$ 1.3°, max 14.2° on R04), and excluding the two tilted frames changes nothing (axis 167.4°, axial R 0.31, hole-at-zero 0.18, median 31.0 m on $n = 608$). The nadir-point correction hypothesis (error $\approx$ alt$\cdot \tan(\text{tilt})$) was tested and killed: a cross-validated correction over all four sign conventions buys 1.6 m against a 2 m bar, and the tilt correlation that motivated it ($+0.44$ with error magnitude) was traced to tilt degrading the homography fit rather than to projection geometry. (b) *Imagery vintage.* Every region is a single-day flight (R03 2018-10-23, R04 2018-10-24, consecutive days in the same district), so any satellite-vintage gap is identical across the pair—yet R03 is the clean control and R04 carries the class. Sec. IV-I adds matched-modal controls showing terrain class operates independently of vintage.

B. The backwards-rate benchmark

Setup. Production matcher, regions R03 (control, 0 fatal), R04 (sub-tile), R06 (whole-tile); drifts 150/300/600 m; 255 pooled accepted fixes. Four rejectors plus two prior-ratio forms.

Finding 1—sequential consistency and the three-keyframe rule discriminate backwards on the sub-tile class. They keep wrong fixes at a *higher* rate than correct ones at every drift (ratio $\leq$0.97 wherever n permits); at tol=100 m, 100% of fatal fixes survive at d150 and d300 (Fig. 3).

Finding 2—the forward ratios that do exist are coverage-destroying. PCM and frame alignment reach ratios of 2–3 at tol=100 m only by keeping 7–40% of correct fixes—the purity-for-coverage trade, measured as a rate. No consistency-based method passes the adoption bar in any cell at any drift.

Finding 3—the whole-tile class is fully separable, the sub-tile class is invisible, per region. On R06, the prior-ratio gate with honest uncertainty keeps 100% of good fixes and 0 of 7 fatal fixes per drift at d150 and d300 (14 fatal fixes pooled over the two drifts). On R04 it scores 0.93–1.00: a 20–80 m alias is small against a 300 m prior, so no ratio threshold can

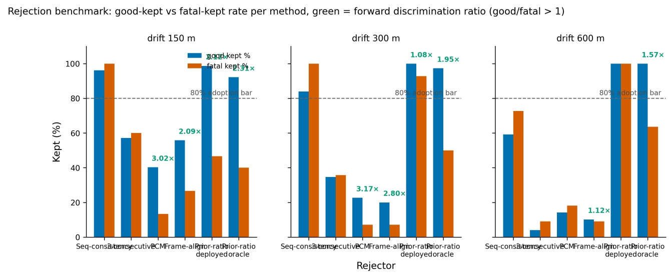

Fig. 3. Backwards-rate benchmark over the pooled fix stream (255 accepted fixes). Bars: good-kept vs. fatal-kept percentage per rejector and drift; dashed line: the 80% good-kept adoption bar. Sequential consistency and the three-keyframe rule keep fatals at $\geq$ good rates (ratio ≤ 1); PCM and frame alignment score >1 only by destroying coverage.

TABLE II
MEDIAN PAIRWISE OFFSET DIFFERENCE VS. FRAME LAG (M). NULL: OFFSETS SHUFFLED WITHIN GROUP (10 TRIALS). n = ALIAS PAIRS.

lag	n	alias: med / null	ratio	good: med / null	ratio
1	25	27.5 / 70.1	**2.55**×	16.7 / 17.9	0.93
2	17	46.8 / 86.1	1.84×	15.9 / 17.9	0.89
3	15	27.9 / 63.9	2.29×	17.2 / 19.3	0.89
5	12	80.0 / 90.1	1.13×	19.6 / 17.3	1.13
8	11	60.3 / 93.5	1.55×	14.6 / 18.6	0.78

see it. On the R03 control, the consistency filters destroy 15–76% of good fixes—collateral measured, not asserted.

C. Temporal coherence: the mechanism

Setup. Truth-referenced signed offsets, contiguous frames, R04 $n = 610$ (alias ≥ 50 m: 93; mid 20–50 m: 374; good <20 m: 143), R03 control $n = 557$. Per-lag pair counts inline.

Finding 4—the alias offset is locally coherent and decays with lag. Adjacent alias frames differ by 27.5 m where random pairing gives 70.1 m; the advantage decays to $\sim 1.1\times$ by lag 5–8 (Fig. 4). The lock is not rigid: it drifts slowly (the period index changes occasionally), coherent over 2–4 frames. Correct fixes are independent per frame (at null at every lag, 0.78–1.13×) and the R03 control matches them. The 20–50 m band is *partially coherent*—a mixture of alias and noise—which explains why threshold rejectors tuned anywhere in that band fail (Finding 1).

Consequence. Within the coherence window, an alias chain is exactly as motion-consistent as the truth: pairwise consistency has a gauge symmetry (constant offsets cancel in frame differences), so PCM/sequential consistency are provably blind, not merely weak. Beyond the window, the offset has drifted enough to survive any tolerance that keeps good fixes (their own ~ 17 m scatter). The backwards rates of Sec. IV-B are the statistical image of this geometry.

D. The countermeasure fails for a measured reason

Setup. Two routes: (a) estimate the alias lattice from the map itself—detrended autocorrelation of a 500-px crop around each locked position, strongest local maximum in the 9.5–95 m band, $n = 610$; (b) a multi-hypothesis mixture filter: candidate true positions on a $k \in [-4, 4]$ lattice (period 20 m, axis 171°,

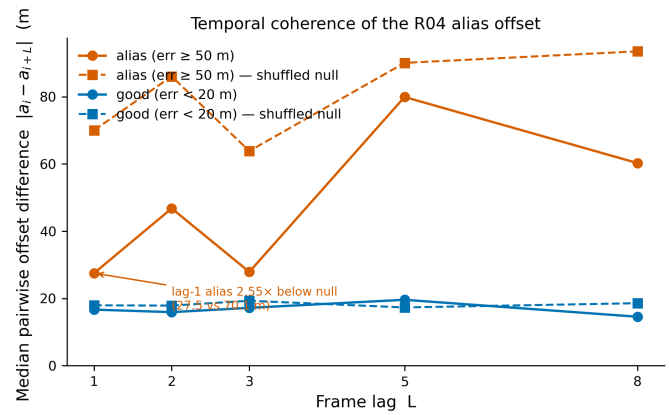

Fig. 4. Temporal coherence of the alias offset (R04, $n = 610$ contiguous). Solid: median pairwise offset difference within the alias (≥ 50 m) and good (<20 m) groups; dashed: shuffled nulls. The alias group sits 2.55× below its null at lag 1 and decays to $\sim 1.1\times$ by lag 5–8; correct fixes are independent at every lag.

from Sec. IV-A), posterior from the prior likelihood, output = posterior mean; prior uncertainty swept 5–300 m in random-walk and IID forms. Kill criterion pre-registered: posterior-mean output must recover $\geq 30\%$ of the oracle gap (17.1 m $\Rightarrow$ ≥ 5.1 m) at prior $\sigma \leq 20$ m.

Result (a)—the lattice is not in the map. In-band periodicity is found on 9–17% of frames (lowest for the alias group) and never aligns with the alias offset direction (median angular error 44–52°, uniform). At 1.19 m/px the satellite texture does not expose the furrow lattice the matcher locks onto.

Result (b)—the mixture recovers nothing; MAP is actively harmful. Baseline $k = 0$ median 31.1 m ($n = 610$); single-axis oracle 14.0 m (gap 17.1 m). Posterior-mean output moves $-6.1 \ldots + 0.1$ m across all prior forms and qualities (worst at $\sigma = 50$ IID: -6.1 m, far below the ≥ 5.1 m bar); MAP output degrades to 68.4 m at $\sigma = 300$ m. The failure is mechanical: frame differences cannot see a constant offset (Sec. IV-C), the prior is the only anchor, and the true offsets are not multiples of any single global lattice—the per-field revision of Sec. IV-A at stream scale.

Finding 5—recovery requires the per-field lattice, which no deployed signal provides. The failure chain is closed: the alias offset is quantized along a field axis (Sec. IV-A), the axis is not in the satellite texture (a), frame differences cannot see a constant offset (Sec. IV-C), and no prior tightness restores the modes (b). The 14.0 m oracle is the non-degenerate bound for any matcher-side fix; the IID $\sigma = 5$ mixture—the video-rate odometry limit—extracts nothing from it (31.2 m).

E. The terrain boundary of the class: exhaustive map

All eleven regions of the dataset are classified with an instrumented GT-tile probe ($n = 40$ step-sampled, DEM-corrected AGL, ≥ 15 MAGSAC inliers, correspondence counts and per-frame tile availability recorded). Seven regions belong to the *unsolvable/ceiling class*; two are clean controls; one carries the whole-tile class; exactly one carries the sub-tile class.

TABLE III
TERRAIN BOUNDARY OF THE ALIAS CLASSES: EXHAUSTIVE 11-REGION MAP. SOLVED = GT-TILE PROBE ($n = 40$ STEP-SAMPLED UNLESS STATED). CORR MED → INLIER MED = MEDIAN CORRESPONDENCE COUNT → MEDIAN MAGSAC INLIERS (CONVERSION %).

Region	Terrain	Solved (GT tile)	Corr→Inl.	Class / dominant cause
R01	riverside/urban	1/40	38→5 (13%)	unsolvable; non-planar
R02	farmland/river	3/40	42→5 (12%)	unsolvable; non-planar
R03	farmland (clean)	27/40	84→33 (39%)	control
R04	furrow farmland	610/738 (83%)	—	**sub-tile aliases**
R05	mountain plateau	0/40	24→5 (21%)	unsolvable; ∼400 m relief
R06	forest	solved	—	whole-tile aliases
R07	suburban hills	0/30	5→4 (80%)	unsolvable; 30-frame flight
R08	suburban/water	3/40	57→5 (9%)	unsolvable; non-planar
R09	suburban	3/40	54→6 (10%)	unsolvable; non-planar + 14/40 GT tiles absent
R10	orchard hills	0/40	16→5 (30%)	unsolvable; relief + canopy
R11	desert	199 frames: 8 alias (4%)	—	control

Three candidate explanations for the ceiling class are ruled out by measurement. (a) *Map quality:* every satellite map is 0.27–0.38 m/px, and per-frame GT tiles exist for all regions except R09 (14/40 missing). (b) *Altitude/scale:* AGL correction is applied from DEM grids (R05 at 250 m spacing; R02/R07/R10 grids built for this revision—raw absolute height vs. ground elevation differences reach 2.7–4.6×); a focal-length sweep (500–1200 px) does not move the inlier ceiling on any region. (c) *Attitude:* median |tilt| is 1.4–2.0° on the failing regions and inliers do not correlate with tilt (+0.02 to +0.15). What the probe shows instead is a correspondence-to-inlier conversion collapse: failing regions still produce 16–57 correspondences—the imagery has features—but only 9–30% convert to geometrically consistent inliers, versus 39% on the working farmland control. The ceiling class is primarily a **geometry** failure: the homography's planar-ground assumption against buildings, river banks, canopy, and mountain relief, not a texture failure. A four-way verification confirms the reading: direct photometric alignment (template NCC with yaw×scale search, ECC refinement) finds no signal on any region—including the working ones, because bulk cross-season/cross-sensor pixels do not correlate (NCC ≈ 0.06 at the known-true alignment of a frame the feature pipeline solves)—so intensity-based alternatives are structurally unsuited; and relaxing the matcher (ratio 0.85, RANSAC 10–15 px) converts the ceiling class from a *no-fix* failure into a *wrong-fix* failure: solve counts rise but the new locks land 107–467 m from truth. The no-fix/wrong-fix boundary is a threshold choice, and the strict side is the safe one. The sub-tile class is therefore doubly specific: it requires not only repetitive solvable terrain but *planar* repetitive terrain—which is why the only furrow farmland in the dataset is also the only place the class appears.

F. External replication: attempted and executed

Attempted replication 1—TEMPO-VINE (rejected on structure). The most promising published agricultural dataset for this mechanism, TEMPO-VINE [9], provides RTK-grade GPS truth (1–2 cm) in trellis and pergola vineyards—but its imagery is a forward-oblique RealSense D435 RGB-D camera on a ground rover at ∼1 m height. Nadir aerial imagery matchable against satellite ortho does not exist in the dataset, so protocol requirement 1 fails and no measurement is possible.

TABLE IV
AERIALVL REPLICATION (PROTOCOL STEPS 1–3).

Metric	Seq 03-11	Seq 03-16
Solved frames (≥15 inliers)	94/200 (47%)	0/200 (0%)
Hole at zero	5.3%	—
Axial resultant $R(180°)$	0.70	—
Anisotropy (along/perp)	9.7× (16.3/1.7 m)	—
Median error	17.1 m	—
Median offset vector	(−16.0 N, +4.1 E) 16.5 m	—
Good-group lag-1 coherence	**0.59× of null**	—
Error groups (good/mid/alias)	75 / 15 / 4	—

This is a *structural* absence in the field: no public dataset currently pairs repetitive-crop aerial imagery with satellite references and continuous GPS.

Executed replication 2—AerialVL (class absent; protocol guard fired). We ported the full protocol to AerialVL, an independently collected UAV dataset (HuggingFace `hmf21/AerialVL`, cc-by-4.0, no paper; per-frame GPS in the image filenames; Qingdao, China; urban/coastal terrain) [36]. Two flight sequences were ingested (1,423 frames), the reference map built from the dataset's georeferenced ortho (0.959 m/px), and protocol Steps 1–3 run.

Two findings follow. First, the alias class is **absent**: the alias tail is 4 frames (4.3%), too thin for any coherence measurement. The sign-folding signatures that *would* replicate (hole 5.3%, R 0.70, anisotropy 9.7×) are produced by a different mechanism—a constant 16.5 m common-mode georeferencing offset between the dataset's ortho and its drone GPS (spread 3.1 m; a cross-validated bias correction removes 14 m: 17.3→3.3 m). A constant offset is trivially "perfectly coherent", so this is a false-positive source for the diagnostic—and a live corroboration of Sec. IV-C's gauge symmetry. Second, the protocol's own control guard fired: the *good* group is temporally coherent (0.59× of its null; a clean dataset requires 0.8–1.25×), meaning this dataset's GPS truth is temporally correlated—a benchmark-hygiene property worth reporting on its own.

Follow-up. The one dataset class that could replicate the mechanism—repetitive-crop aerial imagery (ViLD, WACV 2026, access-gated [37])—is stated as future work with the pre-registered protocol.

TABLE V
SIGNATURE REPRODUCTION ACROSS MATCHERS (GT-TILE, R04). SAME AXIS, HOLE AT ZERO, COHERENT ALIAS TAIL UNDER EVERY FAMILY.

Matcher	Solved	axial R	axis	hole	med. err	lag-1
pooled	610 (83%)	0.31	167.2°	0.18	31.1 m	2.55× (n=25)
ORB-only	497 (67%)	0.28	169.3°	0.19	31.1 m	2.09× (n=14)
SIFT-only	469 (64%)	0.28	166.5°	0.19	30.0 m	2.24× (n=10)
SP+LightGlue	377 (51%)	0.23	167.4°	0.22	30.2 m	1.57× (n=24)

TABLE VI
RATE REPRODUCTION (R04, D300, TOL/τ=100). CELLS: GOOD%/FATAL% (RATIO); DENOMINATORS INLINE. LIGHTGLUE CELLS UNDERPOWERED (2 FATAL FRAMES).

Rejector	pooled (27g/7f)	ORB (22g/7f)	SIFT (23g/8f)	LG (18g/2f)
Seq. consistency	85/100 (0.85)	91/100 (0.91)	78/100 (0.78)	83/50 (1.67)
3-consecutive	30/57 (0.52)	27/43 (0.64)	26/25 (1.04)	22/0
PCM max-clique	19/14 (1.30)	23/14 (1.59)	22/12 (1.74)	22/0
Frame alignment	26/14 (1.81)	32/14 (2.23)	30/12 (2.43)	22/0
Prior-ratio (oracle)	93/100 (0.93)	91/100 (0.91)	91/100 (0.91)	89/100 (0.89)

G. Matcher independence

Setup. The Sec. IV-A signature protocol and the Sec. IV-B rate protocol re-run with the matcher variants on R04 (contiguous GT-tile stream, 738 attempted) and the d300 fix stream ($n = 40$, seed 1992).

Every matcher family reproduces the signature: the same axis within 3°, the hole at zero, the coherent alias tail (1.57–2.55× below null), and good fixes at null. The sub-tile class is a property of the terrain, not of the descriptor pipeline. Solve rates differ (83%→51%), and the alias fraction among solved frames is stable (15–17%).

Finding 6—the rate structure is matcher-stable. Under every classical matcher, sequential consistency discriminates backwards (0.78–0.91), PCM and frame alignment go forward only by keeping 19–32% of good fixes, and the sub-tile class remains invisible to the prior-ratio gate (0.89–0.93) while staying fully separable on whole-tile R06. The LightGlue rate cells are underpowered and reported as such: its accepted-fix stream carries only 2 fatal frames (the learned matcher plus NCC-0.30 acceptance collapses yield—R03 control 9/40 matched vs. 34/40 pooled), so LightGlue's evidence is at the signature level; its one nominal adoption-bar pass rests on $n = 2$ fatal and cannot carry a decision under the small-cell rule.

Dense matchers—Result 7: a third failure mode. RoMa (`roma_outdoor` [26]) is run on the GT-tile protocol across all eleven regions ($n = 167$ frames total). Two discriminators are applied per frame: the implied-translation field of the confident dense correspondences (alias lock ⇒ coherent constant shift; periphery dominance ⇒ uniform spread with centre-anchored H), and gradient-magnitude NCC at the dense lock versus the correct sparse lock. (A reproducibility note: an earlier probe of the same model reported 569–1684 m errors; the position recovery applied the inverse homography to a drone-space point. The corrected run uses the forward homography and is verified against the production recovery on the same frames—4.3 m vs. 893.8 m on one frame—so all dense-matcher numbers here are from the corrected run.)

TABLE VII
SPARSE POOL (CONTROL) VS. ROMA DENSE ON THE GT-TILE PROTOCOL.

Region	ORB pool (control)	RoMa dense
R03 (control)	A@25 50%, 4–23 m	0% A@50; 85–498 m
R11 (clean)	A@25 27%, 15–78 m	0% A@25; 118–912 m
R01/R02/R04/R06	0–85% per-region	0% A@50 everywhere
R05/R07–R10	per Sec. IV-E	0% A@50 everywhere

RoMa produces 700–2000 MAGSAC inliers with tight reprojection RMSE on every frame and 0% A@50 across all eleven regions—including the two regions the sparse pool solves (R03, R11). The failure is *not* alias locking: gradient NCC at the dense lock never beats the correct sparse lock (0/20 frames); and it is *not* periphery dominance: correspondences are uniformly spread (centre-restricted H fit leaves A@25 at 0%), and the implied translation is a coherent displacement of 55 m–11 km whose magnitude has no projection-geometry explanation (Mercator scale variation across a 774 m patch is ≈0.06 m). The mechanism is *dense-prior hallucination*: with weak cross-season local evidence everywhere, the smoothness prior of the learned flow produces a globally consistent, confidently wrong warp.

Result 8—published dense-matcher accuracies presume modal proximity. The same matcher family reports the strongest fine-localization numbers in the field on contemporary matched-modal imagery: XIAN-VisLoc Table 14 gives RoMav2 15.07 m mean error at 95.24% success against 27.24 m at 63.10% for SuperPoint+LightGlue on the same task class [16]. On the vintage-mismatched pairs of the source dataset—same task, same geometry, 0–6 year imagery gap—the family scores 0% A@50 with the largest inlier counts of any matcher. The ordering between sparse and dense matchers is therefore a property of the imagery regime, not of the matcher. Concurrent work converges on the same regime dependence from two other directions: LoRetta [17] documents that direct dense correspondence prediction is unreliable across acquisition-time and season gaps at remote-sensing scale, and Sat-RoMa [18] re-engineers RoMa with a satellite-tuned DINOv3 encoder specifically to survive multi-temporal UAV-to-orthophoto pairs. Our contribution here is the isolated mechanism in the stock models—not alias locking, not periphery dominance, but dense-prior hallucination, with the discriminators that separate the three—and the matched-modal contrast that explains when the published numbers hold.

H. The field's standard fixes, measured

PnP against the DSM on the ceiling class—killed. Lifting GT-tile correspondences to DEM elevation (90 m grid) in the AnyVisLoc/OrthoLoC/OrthoTrack geometry and solving PnP-RANSAC on the seven ceiling regions solves *fewer* frames than the production homography everywhere (healthy control R03: 32 vs. 12 of 40) and worse where it solves (median 34–338 m vs. homography's 11.6–37.6 m). The ceiling class is correspondence-limited, not model-limited.

GNC graduated robust frame alignment [14]—killed. The graduated-robustness form of the Sec. IV-B frame-alignment model keeps 0–7% of fixes in every cell under every matcher at every drift: the μ-graduation converges to an empty inlier set because the prior$\leftrightarrow$estimate offset has no constant mode (Sec. IV-C's scatter). GNC is the sixth rejection family measured, and its behaviour is the cleanest confirmation of the gauge argument: graduated robustness does not rescue the class, it refuses to believe any of the stream.

Robust fixed-lag smoother (GTSAM/TACO-class)—killed with mechanism. A factor graph with GM-kernel map-fix factors and trusted prior-delta motion factors (implementation validated on synthetic data: an isolated 199 m wrong fix is pulled to 8.9 m) does not heal either alias class on real streams. At d150 it moves R06 whole-tile aliases a median 367 m yet lands them wrong—alias chains follow the aircraft, so the trusted motion chain encodes them. At d300 and d600 the smoother degrades sharply (R03 13.9$\rightarrow$54.5 m median; R04 30.9$\rightarrow$142.4 m) because at 7 s frame spacing the drift random walk dominates the prior deltas. A robust back-end needs trusted odometry this dataset does not provide—measured rather than assumed. This result also bounds the deployed cousins: NaviLoc's MAP smoothing with outlier clamping [4] operates at 9 FPS with VIO-grade relative motion—the trusted-odometry regime—and our measurements say that regime is not a convenience but a requirement; at coarse spacing the same architecture's assumptions invert.

Further standard remedies, measured and closed. Four additional published-style remedies were implemented with pre-registered kill criteria. (a) *Zero-shot retrieval floor* (DINOv2-small global descriptors over a 5,234-tile index): the GT tile ranks 53–1556 of 5234 on the control region, and top-5 oracle median error spans 250 m–950 km per region. (b) *Detector fine-tuning* (homographic adaptation of SuperPoint on 80 matched pairs): the pretrained detector is already converged (cross-entropy floor 0.058, 90.5% label self-consistency); the zero-solve behaviour of SuperPoint+LightGlue on frames the ORB pool solves (0/8 held-out control) localizes the gap to the LightGlue matcher, not the detector. (c) *Per-terrain acceptance thresholds* (R09 NCC 0.30$\rightarrow$0.10): match rate 2.5$\rightarrow$7.5% but the extra solves include a 91 m fatal. (d) *Terrain-driven fix weighting* (DEM elevation variance as an adaptive covariance multiplier, Yao-class [27]): DEM variance does not predict match-rate ceilings ($r = -0.21$ over eight regions), and image-moment frame preselection carries no solvability signal (test-half AUC 0.52).

None of the field's standard remedies rescues the ceiling or alias classes on this data; the boundary map of Sec. IV-E is a property of the reference imagery and the frame spacing, not of the estimator choices.

I. Spacing isolation: fix propagation at video rate

Setup. The Sec. III-E propagation chain is evaluated on XIAN-VisLoc [16], an independently collected video-rate dataset (DJI, nadir camera, Level-19 Google reference at 0.24 m/px, per-frame GPS). Trajectory 16 (429 frames, 490×490, 3.8 m median frame step) hosts the controlled pair; the seed uses a truth-centred crop, matching the flight-start GPS assumption. Two arms differ only in frame stride: dense (stride 1) and coarse (stride 65 $\approx$250 m—the source dataset's 7 s regime). The per-frame baseline (independent match per frame on a truth-centred crop) is measured on the same frames for reference.

TABLE VIII
SPACING ISOLATION ON XIAN-VISLOC TRAJECTORY 16 (429 FRAMES). THE TWO PROPAGATION ARMS DIFFER ONLY IN STRIDE.

Arm	Fixed	yield@50m	p50	modes (s/p/r/c)
per-frame baseline	317/429	—	19.7 m	74% solved
prop. dense (3.8 m)	**418/429**	**100%**	**21.1 m**	1/367/50/11
prop. coarse (250 m)	4/7	25%	275 m	1/3/0/3

Finding 9—temporal spacing is the governing variable. At metre-scale spacing the chain carries one anchored fix for the entire flight: 418 of 429 frames fixed (97%), 100% within 50 m, p50 21.1 m—equal to the per-frame baseline (19.7 m) at zero accuracy cost, with satellite re-anchoring reduced to 50 frames total. At 250 m stride the identical chain collapses (4/7 fixed, p50 275 m): KLT track survival across a 250 m gap is the failure point, the same mechanism that kills every consistency-based and smoothing-based remedy on the source dataset (Sec. IV-B, IV-H). The sub-tile alias class is a property of the 7 s capture regime, not of any estimator: the paper's negative results and this positive result are two arms of one variable. The chain architecture is the one fielded by OrthoTrack at video rate [2] and assumed by NaviLoc's VIO-anchored smoothing [4]; what this experiment adds is the controlled isolation—identical chain, stride the only variable—placing the regime boundary between 3.8 m and 250 m spacing, with the collapse mechanism (track survival, not accuracy) identified.

Finding 10—terrain class operates independently of imagery vintage. The per-frame baseline on 14 trajectories of the matched-modal dataset (contemporary satellite, no vintage gap) spans 2–88% solvability:

TABLE IX
MATCHED-MODAL PER-FRAME BASELINE (CONTEMPORARY IMAGERY, CROP AT GPS), 14 TRAJECTORIES.

Solve band	Trajectories
2–19%	Xian04 2%, Weinan01 12%, Xian01 16%, Xian06 19%
31–39%	Xian03 31%, Xian02 39%
52–64%	Xian05 54% (p50 8.6 m), Xian08 55%, Xian09 52%, Xian10 64%
72–88%	Xian11 72%, Xian16 74%, Xian12 84%, Xian07 88%

With contemporary imagery, a quarter of trajectories still land at the 2–19% ceiling-class rates—high-altitude homogeneous terrain (Weinan01, 350 m AGL; crop-size control rules out protocol artifacts) fails exactly like the source dataset's ceiling regions. Terrain class and imagery vintage are orthogonal factors: vintage explains part of the unsolvable-class rate, terrain explains the rest, and no trajectory on either dataset escapes the taxonomy of Sec. IV-E.

Finding 11—the constructive design follows from the regime. A system that (i) anchors once at flight start (GPS

TABLE X
FIX AVERAGING OVER k ACCEPTED FIXES (POOLED D300; BOUND, TRUTH REMOVED PER FRAME).

	k=1	k=3	k=5	k=10
Pooled CEP50	21.8 m	16.7 m	13.8 m	8.0 m
Pooled fatal50	15.7%	12.0%	9.1%	4.8%
R06 CEP50	23.3 m	26.0 m	42.8 m	24.9 m

available), (ii) propagates via geo-registered tracking at video rate, (iii) re-anchors only on track loss, and (iv) coasts on odometry when re-anchoring fails turns the failure taxonomy into a deployment recipe: 97% frame coverage at baseline accuracy on matchable terrain, bounded drift growth elsewhere. The matcher moves from a per-frame duty (0.5–7.4 s/frame in the production pipeline) to a rare re-anchor—the regime change the taxonomy implies, and the regime the deployed systems [2], [4], [5] already operate in.

J. Fix averaging helps—except where aliases cohere

Setup. The pooled accepted-fix streams of Sec. IV-B (R03/R04/R06, three drifts) fused by inverse-variance weighting, weight = inlier count—the production adaptive-R mapping—with per-frame truth removed first. Read this as a bound on extractable signal, not a filter: motion between 7 s frames is unknown, so no navigation filter can access exactly this average. If even the bound fails, averaging is dead; where it wins, an odometry-gated filter may reach it.

Finding 12—averaging is non-monotone on coherent terrain. Pooled and on R03/R04 the median falls monotonically, reaching 8.0 m at k=10—the Report-8 curve shape reproduced under honest protocol (21.8→8.0 against the original 32.4→9). On R06 it climbs first: k=5 (42.8 m) nearly doubles k=1 before the good fixes outvote the lock at k=10. Small averages of a coherent-alias stream concentrate the bias. Same mechanism as Finding U, in fusion form. The consequence is narrow but firm: fuse inside odometry-gated windows (deployment has VIO; Sec. IV-I), never blind short averages on whole-tile terrain.

V. DISCUSSION

A. What to deploy now, and what would actually fix it

The measurements are negative for one specific tool—consistency-based rejection on the sub-tile class—and positive for several others:

1) **Deploy the prior-ratio gate for the whole-tile class.** With honest prior uncertainty it keeps 100% of good fixes and rejects every whole-tile fatal fix at deployment-relevant drift (0/7 per drift, d150 and d300). It requires no matcher change. Operating curve since measured: thresholds 1.5–2.0 hold across drifts; honest filter-reported uncertainty works at drift 150 (all R06 fatals gone, 94% of good kept) and fades by 300, so effectiveness is uncertainty calibration, not threshold tuning. Miscalibration fails open—inflated uncertainty passes everything, keeping 100% of good fixes at thresholds ≥1.5 under oracle, honest, and 3–5×-inflated uncertainty alike—never a good-fix eater. R04 ignores the gate at every setting.
2) **Do not deploy consistency filters as defaults on this task.** They fail backwards or neutrally on the sub-tile class (Sec. IV-B, IV-G) and destroy 15–76% of good fixes on healthy terrain.
3) **Report signed offsets, and run the truth-correlation guard.** The sign-folding diagnostic is what exposed the alias class here and a real 16.5 m georeferencing bias in AerialVL; the guard costs one shuffle test per dataset.
4) **Correct AGL scale before anything else.** The DEM/AGL correction is the difference between a matcher that cannot run on elevated terrain (R06: 10% match) and one that can (50%)—a prerequisite for every number in this paper.
5) **Expect the ceiling class to be geometry, not texture—and do not buy dense matchers to fix it.** Six of eleven regions fail the planar-homography assumption with abundant features (16–57 correspondences, 9–30% geometric conversion), and the strongest dense matcher fails every one of them with the largest inlier counts on record. Where dense matching *is* wanted across a vintage gap, use the adapted variants [17], [18], not the stock models.
6) **Capture at video rate and propagate (Sec. IV-I).** The single regime change that measurably contains the failure classes: anchor once at flight start, propagate the fix by geo-registered tracking, re-anchor only on track loss, coast on odometry. Measured: 74%→97% frame coverage at zero accuracy cost on 429 video-rate frames. Per-frame matching at 7 s spacing—the setting under which every negative result in this paper was produced—is the anti-pattern.
7) **Know where the difficulty is not.** Tile-crop-to-tile matching on Bhopal-terrain reference imagery (40 points, 300 m drifted priors, production ORB stage, no perspective or vintage gap by construction) solves 40/40 at 0.9 m median with a median 340 inliers. Same-modal matching is near-trivially easy—perspective gap, vintage gap, and retrieval carry essentially all real difficulty. Effort spent on the matcher stage past the production configuration buys nothing; effort belongs in the gaps above.
8) **Credit the estimator, with the receipt.** The weak 2D Kalman replay diverges to 215.7 m mean on σ=15 m fixes where the full ESKF (bias tracking, Huber, vision velocity) holds 12.5 m on the same fix class—estimator machinery carries ∼17×. Corollary from the same runs: the production adaptive-R formula overstates noise ∼40× on easy matches (inliers in the hundreds)—conservative exactly where overconfidence would hurt. Safe direction, keep it.

What would fix the sub-tile class itself, with measured distances: (a) *per-field lattice annotation* of reference imagery—the only signal that contains the axis; the oracle bound says 17.1 m of median error is recoverable if the axis is known

(31.1→14.0 m). (b) *Video-rate capture*—Sec. IV-I shows the video-rate regime is the fix for availability, though the alias offset within a matched lock remains per-frame unrecoverable (Finding 5 is unchanged). (c) *RTK-grade truth* to establish whether the ∼13 m R03 floor is the dataset's own GPS noise. Two classical matchers now agree on the floor independent of the production pipeline (ORB-only 11.4 m, SIFT-only 11.7 m vs 13.9 m at drift 300; LightGlue 15.0 m, underpowered at n=9)—not matcher-side—and contemporary farmland (Xian05, 8.6 m) sits below it—not farmland physics. RTK remains the closer. These are design targets with quantified distances, not open wishes.

B. What the field should take

(i) Pooled A@X m metrics on repetitive terrain hide a failure class with full geometric support; signed-offset reporting is a cheap diagnostic that exposes it. (ii) Robust-estimation defaults (PCM/GNC/sequential consistency) do not transfer to cross-view geo-localization on repetitive terrain: they fail backwards or at coverage-destroying purity, at measured rates, under every matcher tested. (iii) The fix that works for one alias class does not exist for the other; hedging (posterior-mean) is safe but inert, mode-commitment (MAP) is dangerous (68.4 m at $\sigma = 300$). (iv) An independent replication attempt shows the diagnostic's false-positive mode and a truth-correlation guard that any future replication must pass. (v) Dense matcher benchmarks transfer only within a modal-proximity regime; failure analyses of learned matchers must state their imagery vintage. (vi) Frame spacing is a first-class experimental variable: the same chain that collapses at 250 m stride delivers 97% frame coverage at metre spacing.

C. Limitations

The sub-tile class is measured on a single region—because the source dataset contains exactly one region that is both solvable and repetitive furrow farmland, as the exhaustive 11-region boundary map establishes rather than assumes; the signature replicates across four matchers, within the flight by split-half, and the external dataset that could replicate the mechanism does not exist publicly. Rates are measured under three classical matchers with a stable structure; the learned matcher is underpowered for rates (2 fatal frames) and supports the claim at the signature level only. Dataset GPS truth (∼10 m class) bounds all absolute numbers—the R03 floor of ∼13 m is plausibly the dataset's own noise floor. AerialVL's georeferencing bias and correlated truth were discovered, not controlled for. The dense-matcher measurements use RoMa v1 (`roma_outdoor`), not the RoMav2 model behind the XIAN-VisLoc contrast, and not the satellite-adapted Sat-RoMa or the matchability-aware LoRetta; the family argument holds, but the specific stronger models were not run—stated, not hidden. The spacing-isolation experiment uses XIAN-VisLoc GPS truth (no RTK), a pure-vision chain (no IMU), and 14 of 21 trajectories for the matched-modal baseline; its accuracy is GPS-noise limited, so sub-20 m claims are untestable on that dataset. The propagation result is a single-trajectory controlled pair; the coarse-arm collapse mechanism is corroborated on the source dataset's seven-second streams. The fix-averaging bound (Sec. IV-J) removes per-frame truth and is unreachable by any filter without odometry; the R03 floor upgrade rests on two classical matchers, the learned one underpowered.

VI. Conclusion

We characterized a failure class in UAV-to-satellite geo-localization that consistency-based robustness tools cannot see: coherent-offset aliases, split into whole-tile (separable by a prior-ratio gate) and sub-tile (unreachable by any measured signal) classes. The sub-tile class defeats sequential consistency and the three-keyframe rule *backwards* at measured rates—under every matcher family tested—defeats PCM and frame alignment through coverage destruction, hides in magnitude histograms, and resists even the mechanism-motivated multi-hypothesis countermeasure, with a non-degenerate oracle bound of 14.0 m median on the worst region against a 31.1 m production baseline ($n = 610$). Dense matchers add a third failure mode—a globally consistent, confidently wrong smooth warp—whose published accuracies transfer only within matched-modal imagery, a regime dependence the field's newest systems now engineer around [17], [18]. The class is terrain-specific in an exhaustively mapped sense: absent on desert, clean farmland, all ceiling-class regions, and an independent urban dataset, and present exactly where terrain is repetitive, solvable farmland. The decisive variable is temporal spacing: at metre-scale frame spacing, geo-registered fix propagation carries one anchored fix for an entire flight (74%→97% frame coverage, zero accuracy cost, 21.1 m median), while the identical chain at 250 m stride collapses—the paper's negative results and its constructive resolution are two arms of one variable, and the deployed systems that assume the video-rate regime now have the measurement that says why they must. The paper contributes the taxonomy, the matcher-stable rate tables, the sign-folding diagnostic, the quantified boundary of what map matching can and cannot recover, and a pre-registered replication protocol for the dataset class the field still lacks. It also contributes the averaging bound with its coherent-terrain exception, and the fail-open characterization of the whole-tile gate.

Data and Code Availability

The measurement harness, rejector benchmark, coherence analysis, and replication scripts, together with all per-frame result artifacts backing every table and figure, are available from the authors; the replication protocol is fully specified (five steps, exact commands, pre-registered kill criteria) so that all experiments can be re-executed end to end. UAV-VisLoc is public [19]; the AerialVL sequences used are public under cc-by-4.0 [36]; XIAN-VisLoc is published with per-frame GPS [16].

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